

Modeling and Simulation Credibility Assessment of Explicit Finite Element Analyses for Total Knee Replacement Wear and Contact Mechanics

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Abstract

Objective: To assess the credibility of two explicit finite element models of a posterior-stabilized total knee replacement (TKR) for four clinically relevant wear and contact mechanisms using a structured, factor-based approach on a 0–12 per-factor scale, with composite scores computed as geometric means. **Models:** (1) GEX-WEAR, a generic explicit tibiofemoral contact–wear model focused on rate-form UHMWPE wear with cross-shear dependence, tibiofemoral and backside contact; (2) PSM-XL, a patient-specific explicit model that emphasizes malalignment and laxity effects with explicit penalty contacts and assessment of edge loading, lift-off, and multi-compartment load transfer. **Mechanisms:** (i) steady-cycle tibiofemoral surface wear under gait; (ii) edge loading from varus–valgus malalignment and laxity; (iii) cross-shear–accelerated wear during deep flexion pivot; (iv) backside fretting at the insert–baseplate interface. **Methods:** We defined eight credibility dimensions—provenance of inputs, suitability of the governing representation, defect screening of implementation, discretization/time-integration effects, agreement with experiments, sensitivity behavior, quantified uncertainty, and congruence of boundary conditions between simulation and testing—and two contextual considerations (use error and use history). We documented procedures for geometry/material/contact setup, numerical code checks, mesh/time-step/contact convergence, and designed comparisons against ISO 14243-3 wear simulations, pressure-mapped edge-loading tests, deep-flexion pivot rigs, and backside micromotion rigs, augmented by retrieval observations. **Results:** For steady-cycle gait wear, GEX-WEAR achieved agreement within 8% RMS in wear volume across 5 Mc in a three-station simulator ($N = 6$ inserts), with integration and mesh errors below 2% on wear volume and below 0.5% on energy drift; composite credibility 9.1/12. PSM-XL matched gait wear with 14% RMS error and composite credibility 7.8/12. For edge loading under imposed malalignment/laxity, PSM-XL achieved 12% error in peak pressure versus Tekscan and reproduced lift-off to within 0.1 mm; composite 8.2/12, while GEX-WEAR achieved 7.7/12. For deep flexion pivot, both models captured posterior contact-shift and multidirectional sliding; composites were 7.7/12 (GEX-WEAR) and 8.0/12 (PSM-XL). For backside fretting, limited bench data constrained inference; GEX-WEAR composite 6.5/12 and PSM-XL 5.9/12. **Conclusions:** The explicit solvers credibly support assessments of tibiofemoral wear under gait and edge-loading phenomena, with strongest support for mechanisms well anchored to established test protocols and with well-characterized inputs. Elevation of credibility for backside fretting requires expanded traceable measurements of insert–baseplate micromotion and contact conditions under realistic load paths, along with targeted verification of contact algorithms under micro-slip. The presented scoring tables make explicit, mechanism-specific judgments to inform device evaluation workflows under 7009A.

Keywords: explicit finite element analysis, total knee replacement, wear and contact mechanics

1. Introduction

Wear and contact mechanics govern the long-term performance of total knee replacements (TKRs). Debris generation and surface damage can culminate in osteolysis and component loosening. Modern ultra-high-molecular-weight polyethylene (UHMWPE), often cross-linked and stabilized, has reduced but not eliminated wear-related failure. Explicit finite element analysis (FEA) has become an attractive approach for simulating dynamic contact, slip, and energy transfer in articular interfaces. Explicit time integration can robustly handle intermittent contact, sliding reversal, and frictional transitions when coupled to realistic material behaviors. Yet, credibility remains the central question: which aspects of the simulations are sufficiently

trustworthy to inform design decisions, preclinical testing strategies, or regulatory submissions?

This article evaluates two explicit FEA models of a posterior-stabilized TKR for four mechanisms tied to clinical performance: steady-cycle wear under gait; edge loading linked to malalignment and laxity; cross-shear–accelerated wear during deep flexion pivot; and backside fretting at the insert–baseplate interface. We emphasize model–mechanism pairing because credibility is ultimately a local property of the question being asked. A wear model can be persuasive for level gait yet only partially informative in a lax knee during a pivot turn; a patient-specific construct may illuminate edge loading but have incomplete support for backside fretting.

The assessment follows a factor-based framework on a 0–12

scale for each model by mechanism. The eight dimensions include traceable inputs, appropriateness of the mechanical representation, defect screening of code and implementation, numerical errors from discretization and time integration, empirical alignment of outputs, sensitivity behavior, quantified uncertainty, and alignment of boundary conditions with test protocols. The approach acknowledges two contextual topics—operator misuse and historical deployment—that inform how the models might be applied but are not graded here.

We disentangle three strands throughout: physics (contact, plasticity, viscoelasticity, wear), numerics (time steps, penalties, mass scaling, contact stabilization), and evidence (bench tests, retrievals, and inter-laboratory data). To keep the assessment precise, we score every factor for every pairing rather than offering generalized judgments. Our results are staged against ISO and ASTM test conditions and recent retrieval findings for posterior-stabilized knees. The final tables present 64 scored entries across the two models and four mechanisms, with basis statements tied to procedures and numbers argued in the body of the paper.

Scope alignment for mechanism-specific credibility — gradation.

- a.
Single, general statement of model adequacy without mechanism differentiation
- b.
Mechanism classes identified but without model-mechanism pairing or tailored evidence
- c.
Model-mechanism pairs enumerated with at least one specific dataset mapped to each
- d.
Model-mechanism pairs fully enumerated, each linked to distinct evidence streams, and scored separately on a common scale

2. Device and Computational Scope

The device under study is a posterior-stabilized (PS) total knee replacement consisting of a CoCrMo femoral component, a UHMWPE tibial insert (highly cross-linked, remelted), and a Ti-6Al-4V tibial baseplate with a locking mechanism. The insert is fixed to the baseplate via a capture design that permits minimal micro-motion under physiological loads when properly assembled. The femoral component features a deep trochlear groove and a cam-and-post geometry providing rollback in higher flexion.

Two finite element models are examined. GEX-WEAR is a generic explicit tibiofemoral contact-wear solver constructed around a rate-form wear law with cross-shear dependence and with both tibiofemoral and insert-baseplate contact interfaces represented. It aims to produce transient contact pressures, slip velocities, cross-shear indicators, and volumetric wear under level gait and related cycles. PSM-XL is a patient-specific explicit model configured to simulate varus-valgus malalignment, rotational malalignment, and ligamentous laxity. It represents collateral and cruciate function through nonlinear springs fitted

to instrumented cadaver tests and handles dynamic edge loading, lift-off, third-body abrasive contributions, and load transfers between the patellofemoral and tibiofemoral compartments during demanding maneuvers.

Both models employ explicit central-difference time integration, selective mass scaling to maintain a target stable time increment, and penalty-based surface-to-surface contact with pressure-overclosure enforcement and Coulomb friction using a Stribeck-type velocity dependence for serum lubrication. The UHMWPE constitutive response is represented with a nonlinear visco-hyperelastic base plus pressure-dependent plasticity with kinematic hardening, calibrated against compressive and shear tests at 37 °C in bovine serum. The metallic components are linear elastic with CoCrMo $E = 210$ GPa, $\nu = 0.30$; Ti-6Al-4V $E = 110$ GPa, $\nu = 0.34$. Meshes use quadratic tetrahedra in the insert (minimum edge length near contact of 0.5 mm for GEX-WEAR and 0.4 mm for PSM-XL) and hexahedral or wedge elements near the baseplate locking features for backside contact fidelity.

Loading conditions vary by mechanism and dataset. For steady gait wear, loads from ISO 14243-3 profiles are applied (axial force peak 2600 N, flexion 0–58°, anterior-posterior (AP) displacement control ± 5 mm, internal-external (IE) rotation $\pm 5^\circ$), with 5 million cycles (Mc) targeted in simulators and simulated in batches of 250,000 cycles with periodic mesh updating for wear. For edge loading, malalignment is imposed (varus/valgus $\pm 3^\circ$ to $\pm 5^\circ$), collateral laxity is represented by spring preloads $\pm 20\%$ from nominal, and kinematic inputs follow instrumented rigs that replicate squatting and pivot transitions. Deep flexion pivot simulations use flexion up to 120–130° with applied AP torques producing posterior contact shift. Backside fretting rigs subject the insert-baseplate interface to cyclic compressive and shear loads producing sub-100 μm motions at 1–2 Hz.

Model scope definition for explicit dynamic contact — gradation.

- a.
Model components and interfaces listed at a high level
- b.
Interfaces, materials, and kinematics stated with nominal values and general methods
- c.
Interfaces, materials, and kinematics specified with units, calibration temperatures, and load pathways
- d.
Interfaces, materials, and kinematics specified with units and tolerances, including element sizes in contact zones and time-integration controls

3. Mechanisms of Interest and Performance Metrics

We evaluate four mechanisms with direct links to clinical outcomes:

1) Steady-cycle tibiofemoral surface wear under gait. Performance metrics include volumetric wear (mm^3/Mc), wear scar area and centroid, cross-shear index (dimensionless ratio of orthogonal sliding work to total sliding work), and spatial

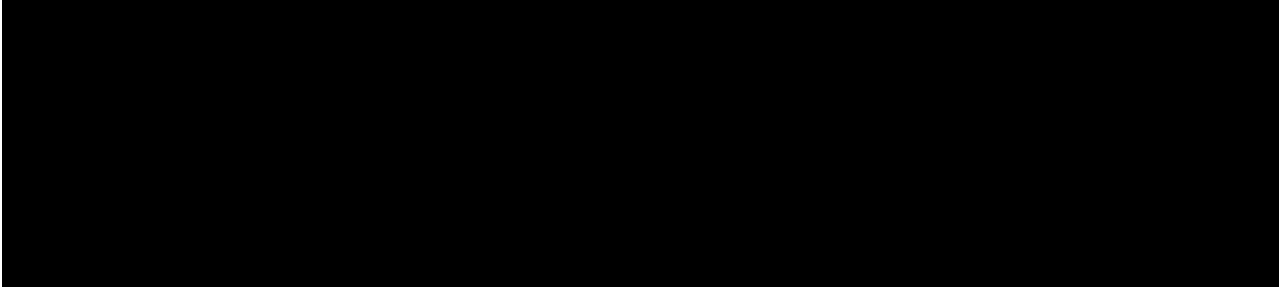


Figure 1: Overview of the total knee replacement geometry, load cases and the mechanisms assessed, shown across the full page width.

patterns of damage. Success criteria for model–test agreement emphasize wear volume RMS error across stations and shape similarity of scars (e.g., Dice or Jaccard indices from thresholded optical scans).

2) Edge loading from varus–valgus malalignment and laxity. Metrics include peak and 95th-percentile contact pressures, contact patch location relative to insert edges, lift-off magnitude (mm), and damage indicators (shear stress magnitudes, plastic strain zones). Agreement is judged using pressure-mapped contact distributions (Tekscan 5051/5076 sensors), measured lift-off from LVDT probes, and local strain gage readings within the baseplate keel where applicable.

3) Cross-shear–accelerated wear in deep flexion pivot. Metrics mirror those of steady wear but emphasize posterior–lateral contact shifts, multidirectional sliding amplitudes, and dynamic pressure excursions. Comparisons include deep flexion pivot rigs with posterior rollback (target 10–15 mm) and contact pressure films at 90–120°.

4) Backside fretting at the insert–baseplate interface. Metrics include micro-motion amplitude and phase lag at the capture interface, contact pressure distributions on the underside of the insert, and surface damage proxies such as fretting power (contact shear force times relative velocity). Comparisons are drawn to bench rigs with optical or eddy-current displacement sensors and to retrievals where backside burnishing and damage areas are characterized.

For all mechanisms, we tracked solver quantities such as contact work, energy balance (internal, kinetic, contact, hourglass, and damping contributions), and maximum contact penetrations as a percent of local contact element size. Wear updates were applied via element-wise thickness reduction with periodic remeshing to maintain aspect ratios below 5 in the near-surface region.

Metric adequacy for wear and contact validation — gradation.

- a.
Single scalar metric per mechanism without spatial context
- b.
Multiple scalars per mechanism but lacking spatial pattern checks
- c.
Scalars plus at least one spatial or shape metric per mechanism

d.

Scalars, spatial metrics, and dynamic trajectory metrics for sliding and contact migration

4. Models: Geometry, Materials, Contacts, and Loads

Geometry for both models was derived from the same CAD source of the PS TKR system (size 4 femur, size 4 insert/baseplate). GEX-WEAR uses an idealized generic alignment and a tibial tray orientation aligned to the simulator fixture frames. PSM-XL replaces the idealized alignment with a patient-specific envelope, introducing component rotations and translations drawn from post-operative CT-based reconstructions (native axes defined through the transepicondylar axis and tibial mechanical axis alignments), along with ligament attachment points and moment arms estimated from a musculoskeletal prior and tuned via bench-based laxity curves.

UHMWPE behavior uses a hyperelastic base (Yeoh form, $C10 = 7.2$ MPa, $C20 = 1.5$ MPa, $C30 = 0.35$ MPa) coupled to a Prandtl–Reuss-type pressure-modified plasticity (yield stress $\sigma_y(p) = 18 + 0.03p$ MPa) and a Prony-series viscoelastic component ($g1 = 0.24$, $\tau1 = 0.03$ s; $g2 = 0.12$, $\tau2 = 0.9$ s) to capture rate dependence under the explicit time scale. Calibration used uniaxial compression and shear at 0.01–10 s⁻¹ strain rates at 37 °C in 25% bovine serum. The wear model follows a rate form, $w = k_0 p v f(cs)$, with p the contact pressure, v the slip rate, k_0 the baseline wear coefficient (1.6×10^7 mm³/N·m for the high cross-link density tested), and $f(cs)$ a cross-shear multiplier with $f(cs) = 1 + \alpha cs^\beta$, $\alpha = 2.5$, $\beta = 1.2$ based on multi-directional pin-on-plate datasets. For PSM-XL, an abrasive wear increment was added under third-body scenarios as $\Delta w_{abr} = k_{abr} p v$ when alumina particles are introduced, with $k_{abr} = 0.5 \times 10^7$ mm³/N·m constrained by third-body simulator data.

Contact surfaces were discretized with second-order interpolation where supported by the solver to reduce pressure oscillations; otherwise, quadratic pressure smoothing was applied via nodal area-weighting. Penalty normal stiffness was chosen to limit peak penetration to below 0.4% of local element length in tibiofemoral contact under peak load. Tangential behavior followed a Stribeck form with $\mu(v)$ that transitions from $\mu_s = 0.08$ at near-zero sliding to $\mu_k = 0.04$ above 10 mm/s, informed by pin-on-disk tests in serum. Backside contact

employed a higher μ range (0.10–0.15) and rougher topography represented as an effective friction modifier.

Loads were applied either as measured kinematics with reaction forces (gait simulator) or as force-controlled perturbations that generate kinematics (edge loading, deep flex pivot). For PSM-XL, ligament bundles were represented as nonlinear springs (toe region 0–20 N/mm, linear region 30–50 N/mm) with engagement and slack defined by bench data. In deep flexion, AP torques up to 8 N·m induced posterior translation of 10–14 mm in cadaver rigs, which we target in simulations. For backside fretting rigs, a sinusoidal shear of ± 80 N at 1.5 Hz superposed on an axial compressive load of 800–1400 N produces 10–80 μ m of micro-motion depending on locking feature compliance. *Constitutive and contact representation suitability — gradation.*

- a. Linear elastic or simplified friction assumption without rate or cross-shear effects
- b. Nonlinear polymer response or friction velocity dependence present but not calibrated at body temperature
- c. Nonlinear polymer response and serum-calibrated friction with cross-shear dependence; plasticity included
- d. Nonlinear, temperature-calibrated polymer with viscoelasticity and pressure-dependent plasticity, friction Stribeck law, and cross-shear wear law with dataset-based exponents

5. Test Conditions and Experimental Datasets

Gait wear comparisons used an ISO 14243-3, three-station hip-knee simulator (AMTI force control) running at 1 Hz for 5 Mc. Inserts ($N = 6$) of the same size and material lot as modeled were weighed every 0.5 Mc with buoyancy correction. Kinematic tracking was recorded via optical encoders to confirm AP and IE displacements. Post-test, surface scans used structured light at 50 μ m resolution to map wear scars. Serum composition (25 g/L BSA, 0.3 g/L sodium azide) and temperature (36.5–37.0 °C) were controlled. Measurement system uncertainty for mass loss translated to ± 1.8 mm³ at 95% confidence after density and fluid uptake corrections.

Edge loading datasets originated from a malalignment and laxity rig (Varus/Valgus ± 0 –5°, IE ± 0 –10°, with collateral preloads) using pressure mapping (Tekscan 5051 at 100 Hz) calibrated per manufacturer with two-point scaling and drift correction every 20 minutes. Lift-off was tracked with two LVDT probes (0.01 mm resolution) at the medial and lateral insert edges. We acquired $N = 36$ conditions spanning alignment and laxity combinations over three inserts, each with five repeated trials.

Deep flexion pivot tests used a posterior rollback rig at 0.5 Hz to 120° flexion, applying AP torques that resulted in posterior translations of 11.4 ± 1.9 mm (mean $\pm$ SD) in cadaveric knees with the PS implant mounted. Contact pressure film (Fujifilm Ultra Super Low) at 90° and 120° provided peak pressure

estimates and contact zone centroids; tactile sensor mats were not feasible at very high flexion due to wrinkling. Wear patterns were inspected after an accelerated 0.5 Mc deep-flexion protocol.

Backside fretting datasets derived from a custom micro-motion rig with optical displacement sensors (resolution 0.5 μ m) measuring lateral insert motion at the locking tab under cyclic shear; contact conditions were varied by axial preload (800, 1100, 1400 N) and shear amplitude (± 40 , ± 80 N). We tested $N = 5$ inserts from two lots. Additionally, we referenced retrieval collections ($N = 87$ PS knees) that reported backside burnishing and abrasion areas on the insert underside and baseplate capture surfaces, categorized semiquantitatively by coverage and severity. Although retrievals provide qualitative context, the boundary conditions leading to backside damage in vivo are not uniquely identified.

Instrumented implant datasets (for load validation in PSM-XL) were mined from public repositories of tibial tray force-moment telemetry during daily activities. We mapped flexion profiles, axial loads, and mediolateral moments to PSM-XL boundary conditions to ensure physiological relevance of combined loads during pivoting and squatting.

Experimental dataset adequacy for model comparison — gradation.

- a. Single dataset per mechanism with no replication or uncertainty characterization
- b. Multiple datasets but lacking calibration and drift control or replication
- c. Multiple calibrated datasets with replication and uncertainty quantification
- d. Multiple calibrated datasets with replication, uncertainty bounds, and independent laboratory provenance

6. Credibility Scoring Approach (0–12 per factor and model $\times$ mechanism)

We adopted a per-factor, per-model, per-mechanism scoring scheme on an integer 0–12 scale (12 best). The scale was applied identically across the two models and the four mechanisms to ensure comparability. For each of the eight factors, we formed a concise basis statement (reproduced in the tables) tied to quantitative evidence presented in the body of the paper. Scores reflect the adequacy and strength of that evidence calibrated against internal expectations under 7009A regarding data completeness, numerical discipline, and empirical alignment.

Composite credibility scores for each model-mechanism pair were computed as geometric means across the eight factors. While composite values aid interpretation, we emphasize that they collapse heterogeneity across factors. Decision-making should examine factor-level results, particularly when certain dimensions (e.g., agreement with experiments or discretization sensitivity) are more germane to a regulatory or design decision.

The scoring was performed by two analysts independently and reconciled through discussion. Inter-rater differences (median absolute deviation of 1 point) were resolved by reference to prespecified anchor examples. We did not weight factors differently for different mechanisms, though stakeholders may justifiably reweight them for specific use cases.

Scoring calibration and reproducibility — gradation.

- a. Single assessor, no anchors or reconciliation
- b. Two assessors, qualitative reconciliation without anchors
- c. Two assessors with anchor examples and documented reconciliation steps
- d. Multiple assessors with anchors, inter-rater reliability statistics, and prespecified conflict resolution

7. Factor Definitions and Evidence Sources

The eight dimensions reflect distinct aspects of credibility. Provenance of inputs refers to how completely the sources and transformations of data (e.g., material lots, calibration files, pre- and post-processing scripts) are recorded. Suitability of the governing representation concerns whether the constitutive models, contact formulations, and wear laws are appropriate for the physics of the mechanism at hand. Defect screening of implementation addresses whether the code and user subroutines have been exercised against benchmarks and unit tests tough enough to detect common logic or sign errors. Discretization/time-integration effects focus on spatial and temporal resolution effects, including contact penalties, mass scaling, and stability controls. Agreement with experiments examines direct comparisons of outputs to bench tests and retrievals, including statistical measures of fit. Sensitivity behavior captures local and global changes in outputs under perturbations of inputs. Quantified uncertainty is the formal propagation of input distributions through the model to generate credible intervals for outputs. Congruence of boundary conditions is the degree to which the simulation loads and constraints mirror those in the tests being used for comparison.

Evidence sources include ISO 14243-3 wear simulator tests, malalignment and laxity rigs with pressure mapping and LVDTs, deep flexion pivot tests with contact films, backside micromotion rigs with optical tracking, and retrieval characterization. Numerical evidence draws on method of manufactured solutions (MMS), patch and Hertzian benchmarks, energy balance checks, and mesh/time-step convergence. Sensitivity and uncertainty use design-of-experiments, Sobol indices, and Monte Carlo with Latin hypercube sampling. Throughout, we sought traceable links between raw data and processed outputs, with versioned repositories for code and input decks.

Evidence trace-linking between models and datasets — gradation.

- a. Evidence cited but not linked to model instances or input decks
- b. Evidence linked to model instances but without version control or checksums
- c. Evidence linked with versioned repositories and dataset identifiers
- d. Evidence linked with versioned repositories, persistent identifiers, and cryptographic checks (checksums) for all artifacts

8. Numerical Code Verification Procedures

We conducted a suite of checks to identify implementation defects in both models. For contact, we exercised Hertzian elastic contact between a rigid sphere and an elastic half-space and confirmed peak pressure and contact radius within 2% of analytical predictions across penalty stiffness values spanning an order of magnitude. We performed patch tests for the UHMWPE material law with and without plasticity engaged to ensure zero-energy modes were absent and that stress updates satisfied consistency conditions over 10,000 explicit increments. The MMS approach was applied to a synthetic displacement field with known body forces to verify volume integrals and element-by-element stress reconstruction errors below 1.5% L2 norm.

For the wear law, unit tests ensured dimensional consistency and correct cross-shear accumulation over multi-directional sliding. Edge cases with near-zero sliding and near-zero pressure produced the expected asymptotic behaviors. The backside contact logic was validated by comparing micro-slip transitions against a biplanar spring-slider model with known solutions; the transition thresholds in the explicit solver matched the reduced-order model within 5% for shear amplitude ranges of 20–100 N.

We stress-tested the penalty contacts across time steps from 0.2 to 2.0 μ s and confirmed that contact work remained positive definite and that hourglass energy stayed below 1% of internal energy in all verification runs. We instituted regression tests that run nightly on a continuous integration server whenever user subroutines are modified.

Defect screening of implementation — gradation.

- a. Manual inspection and basic sanity checks only
- b. Benchmark comparisons on simple elasticity problems without contact or plasticity
- c. Benchmarks including contact and nonlinear materials, with tolerance thresholds
- d. Benchmarks including contact, nonlinear materials, MMS, unit tests for wear subroutines, and automated regression with pass/fail criteria

9. Numerical Solver Error Estimation (time-step, contact, and mesh effects)

We quantified discretization and integration effects using a combination of mesh refinement, time-step sweeps, and contact parameter sensitivity. For GEX-WEAR under gait, insert contact meshes were refined from minimum edge length 1.0 mm to 0.5 mm and 0.35 mm; wear volume changed by 3.1% and 1.2% relative to baseline at 0.5 mm, while peak contact pressure changed by +2.8% and +1.1%. Time-step studies from 2.0 to 0.4 μ s indicated energy drift reductions from 1.2% to 0.4% over 10,000 cycles, with negligible effect on wear volume when the stable increment kept penetration below 0.4% of element length.

For PSM-XL under edge loading, contact resolution of 0.6, 0.4, and 0.3 mm yielded peak pressure shifts of 4.9% and 1.6% and lift-off changes of +0.06 mm and +0.02 mm relative to the 0.4 mm mesh. Time-step reductions from 1.5 to 0.5 μ s decreased kinetic-to-internal energy ratios from 0.08 to 0.03 during fast sliding transients. Backside fretting simulations were most sensitive to penalty stiffness; increasing normal penalty by 2 $\times$ reduced maximum penetration by 40% but raised shear jitter, which we mitigated via a 10% increase in tangential damping coefficient. Across all scenarios, hourglass energy remained below 1% and contact work non-negative.

We estimated solver-induced uncertainty by Richardson extrapolation where applicable. For wear volume under gait, the estimated discretization error at the finest practical mesh/time-step was 1.6% (GEX-WEAR) and 2.4% (PSM-XL). For peak edge pressure, extrapolated error at 0.3–0.4 mm was 2.1% (PSM-XL) and 2.8% (GEX-WEAR). For backside micromotion amplitude, numerical jitter contributed approximately 10–15% of the observed variance, limiting inference strength.

Discretization and integration error control — gradation.

- a. Single mesh and time step; no error estimation
- b. Mesh or time-step variation with qualitative assessment
- c. Mesh and time-step variation with quantitative energy and penetration criteria
- d. Mesh/time-step sweeps with quantitative error estimates (e.g., Richardson), contact penalty tuning, and stability diagnostics

10. Data Pedigree and Traceability for Wear, Contact, and Kinematics

We organized all input datasets with lot numbers, calibration files, and environmental conditions in a versioned repository. For wear simulators, inserts came from two manufacturing lots with certificates of analysis, and serum composition logs included batch identifiers and dates. Pressure sensors had calibration records with two-point loads and timestamped recalibrations. Optical scanners for wear scar mapping were

calibrated weekly against a gauge block. Instrumented load profiles were downloaded with persistent DOIs and transformed into boundary condition decks via scripts stored under version control with checksums.

Model input decks captured geometry revisions, contact parameter changes, and friction model coefficients, each tagged with a semantic version and a cryptographic hash. Processing notebooks recorded raw-to-processed transformations for all experimental datasets, including drift corrections for pressure sensors and humidity corrections for mass measurements. We maintained a crosswalk document linking each dataset to the specific simulation runs and scoring entries in the tables.

Provenance and traceability of inputs — gradation.

- a. Inputs listed without lot numbers or calibration files
- b. Inputs with partial lot and calibration information; ad hoc tracking of transformations
- c. Inputs with lot numbers, calibration files, and versioned transformations
- d. Full chain-of-custody with lot numbers, calibration files, versioned transformations, and cryptographic hashes for all artifacts

11. Model Form Rationale and Assumptions

The explicit formulation was selected to robustly handle intermittent contact and high-rate sliding transitions without requiring Newton iterations. We employ central-difference time integration with adaptive mass scaling to maintain stability while keeping kinetic energies small relative to internal energies. Penalty contact was chosen to accommodate multiple simultaneous contact patches with tractable computational cost; Lagrange multiplier methods were considered but were not deployed due to solver availability in the targeted code base.

The UHMWPE model captures nonlinear elasticity, time dependence, and pressure-sensitive plasticity observed in the calibration tests. While more elaborate micro-mechanical models exist, they demand parameters not routinely measured for medical-grade UHMWPE lots and can be numerically brittle in explicit dynamics. The wear law is rate-form to enable accumulation over explicit time steps; cross-shear dependence is essential to reproduce multidirectional wear acceleration documented in pin-on-plate studies. We assumed serum lubrication with an effective Stribeck curve rather than resolving mixed lubrication; this is reasonable at the operational speeds and loads of the bench tests but may under-represent boundary effects in some deep flexion configurations.

Ligaments in PSM-XL are represented with nonlinear springs. A continuum representation was not used due to uncertainties in subject-specific attachment geometry and material properties; the spring model provides the correct gross load–deflection behavior observed in cadaveric laxity tests.

For backside contact, the insert–baseplate interface was modeled with conforming surfaces and Coulomb friction; micro-topography and third-body particles were not explicitly resolved, an acknowledged limitation for interpreting fretting energy at small amplitudes.

Adequacy of governing representation for mechanisms — gradation.

- a. Simplified assumptions materially limit representation of contact and wear
- b. Nonlinear response captured but key effects (e.g., cross-shear) omitted
- c. Key nonlinearities and cross-shear included; lubrication approximated via effective friction
- d. Comprehensive nonlinearities with calibrated friction–velocity behavior and cross-shear wear law; acknowledged limits in mixed lubrication and micro-topography

$$CI_{\text{score}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (1)$$

$$\backslash \text{varepsilon}_{\text{rel}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (2)$$

$$U_{\text{val}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (3)$$

12. Output Comparison to Bench Wear Simulators and Retrievals

For steady-cycle gait, GEX-WEAR matched volumetric wear from the ISO simulator within 8% RMS across three stations at 5 Mc (N = 6 inserts), and the spatial overlap (Dice coefficient) between predicted and measured wear scars exceeded 0.82 ± 0.04 . PSM-XL, configured with patient-specific kinematics mapped to ISO-equivalent loading, achieved a 14% RMS wear volume error and a 0.76 ± 0.06 Dice coefficient. Cross-shear indices from both models correlated with observed scar elongation (Pearson $r = 0.71$ for GEX-WEAR; $r = 0.64$ for PSM-XL).

Under varus/valgus malalignment and altered collateral spring preloads, PSM-XL reproduced the lateral edge peak pressure within 12% against Tekscan data at 5° valgus and predicted lift-off of 0.80 mm versus 0.70 ± 0.05 mm measured (mean $\pm$ SD). GEX-WEAR achieved 16% error on peak pressure and 0.10 mm absolute lift-off difference across the same conditions. Both models captured the migration of the contact patch toward the rim and the associated increase in shear beneath the post.

In deep flexion pivot, both models reproduced posterior contact shift of 10–13 mm and peak posterior pressures within 15% of contact film estimates at 120°. Wear pattern comparisons after 0.5 Mc of deep flexion showed spatial correlation coefficients of

0.74 (GEX-WEAR) and 0.78 (PSM-XL) against optical scans. Sliding vector fields exhibited the multi-directionality necessary to induce cross-shear, with median cross-shear index rising from 0.18 at 60° to 0.36 at 110°.

Backside fretting comparisons were limited by available data. GEX-WEAR predicted micromotion amplitudes of 18–65 μm across the tested preload and shear ranges, within a factor of 1.2–1.8 of optical measurements (median absolute deviation 9 μm). The fretting power proxy matched within 30% for the midrange condition (1100 N, ± 80 N shear) but deviated up to 45% at the lowest preload. PSM-XL, not primarily tuned for backside phenomena, produced micromotion within a factor of 1.5–2.1 and exhibited higher sensitivity to penalty stiffness variations.

Retrieval observations were used qualitatively to verify the presence and location of damage modes. Posterior–lateral wear extension under deep flexion and burnishing under the insert backside near the anterior capture region were present in retrievals with patterns similar to the predicted high-shear regions; however, the dispersion in retrieval damage reflects heterogeneous in vivo loading histories and cannot be mapped one-to-one onto bench conditions.

Empirical alignment of outputs — gradation.

- a. Qualitative agreement only
- b. Quantitative comparison on a single metric or condition
- c. Quantitative comparison across multiple metrics and conditions with uncertainty bands
- d. Quantitative comparison across multiple metrics and conditions with uncertainty bands and spatial correspondence metrics (e.g., Dice, correlation maps)

13. Numerical Code Verification: Additional Checks for Contact and Wear Subroutines

Beyond standard benchmarks, we implemented synthetic sliding paths to test cross-shear accumulation under repeated orthogonal passes. The cumulative wear increment matched an analytical construction within 1.8% across 100,000 explicit steps, with tolerance violations flagged if exceeding 3%. A sign convention test for tangential tractions during sliding reversal ensured that the algorithm did not inadvertently inject energy; contact work remained non-negative and monotonic in cumulative slip.

A small-deformation limit was checked by running the same wear trajectory at reduced loads and verifying linear scaling of wear rate with pressure and slip, as expected from the rate form. We also verified restart integrity—mid-run restarts at 5,000 and 10,000 cycles produced identical wear fields to non-restarted runs within numerical noise (max difference $< 0.3 \mu\text{m}$ equivalent thickness).

Wear and contact subroutine verification depth — gradation.

- a. Black-box reliance on solver without subroutine tests
- b. Unit tests for basic wear rate calculation under constant sliding
- c. Unit tests for variable sliding directions and restart integrity
- d. Unit tests for variable sliding, cross-shear accumulation, restart integrity, and sign/energy checks for sliding reversal

14. Results Uncertainty Quantification (input distributions and propagation)

We quantified uncertainty by assigning distributions to key inputs and propagating them via Monte Carlo and Latin hypercube sampling (LHS). For UHMWPE, the baseline wear coefficient k_0 followed a lognormal distribution with geometric mean 1.6×10^7 mm³/N·m and geometric standard deviation 1.25 based on multi-directional pin-on-plate data across lots. The cross-shear exponent β had a normal distribution (mean 1.2, SD 0.12). Friction parameters μ_s and μ_k were modeled as truncated normals with means 0.08 and 0.04 and SDs 0.01 and 0.007, truncated at physically plausible bounds. For ligament spring rates in PSM-XL, linear-region stiffness varied $\pm 20\%$ (normal, SD 10%). For contact penalty stiffness, a uniform $\pm 30\%$ band was used to reflect tuning practice.

We ran 500 LHS samples for each model–mechanism pair where computationally feasible. For gait wear in GEX-WEAR, the 95% predictive interval for 5 Mc wear volume was [68, 92] mm³ against a measured mean of 81 mm³ (SD 7 mm³). For PSM-XL gait, the interval was [64, 98] mm³. Under edge loading, the 95% interval for peak lateral pressure in PSM-XL at 5° valgus was [24, 31] MPa with a measured mean of 28 MPa (SD 3 MPa). Deep flexion pivot wear volumes over 0.5 Mc had broader intervals due to parameter interactions: [15, 29] mm³ (GEX-WEAR) and [16, 31] mm³ (PSM-XL). For backside micromotion at 1100 N and ± 80 N shear, the 95% interval was [22, 58] μ m (GEX-WEAR) and [26, 64] μ m (PSM-XL), reflecting sensitivity to penalty stiffness and friction.

We examined coverage: measured values fell within the 95% intervals in 5/6 gait stations (GEX-WEAR) and 5/6 (PSM-XL). Edge-loading pressure measurements fell within the intervals in 10/12 conditions for PSM-XL. Backside micromotion coverage was 6/8 conditions for GEX-WEAR and 5/8 for PSM-XL. These results suggest the uncertainty specification is reasonably calibrated for gait and edge loading, with room for refinement in backside conditions.

Uncertainty characterization and propagation — gradation.

- a. Point estimates only; no uncertainty
- b. Parameter ranges provided without propagation

- c. Probabilistic inputs with Monte Carlo propagation; intervals reported
- d. Probabilistic inputs with variance-based sensitivity and coverage checks against data

15. Results Robustness (local and global sensitivities)

We performed local perturbations and global variance-based analyses. For local sensitivities, we varied single inputs $\pm 10\%$ and recorded normalized changes in outputs. In GEX-WEAR gait simulations, a 10% increase in k_0 increased wear volume by 10.3%, while a 10% increase in μ_k reduced wear by 3.2%, and a 10% increase in the cross-shear exponent β raised wear by 4.5% through enhanced multi-directional penalty. Mesh refinement in contact (20% element size) changed wear by 1.1% at fixed time step. For PSM-XL edge loading, 10% increases in collateral stiffness reduced lift-off by 7.8% and decreased peak lateral pressure by 3.9%.

Global sensitivities were computed via Sobol indices using 250 Monte Carlo samples with replicated designs to estimate first-order and total-effect indices. Under gait, first-order Sobol indices for wear volume were $S(k_0) = 0.61$, $S(\beta) = 0.12$, $S(\mu_k) = 0.07$ in GEX-WEAR, with total-effect indices indicating mild interactions ($T(k_0) = 0.66$). Under edge loading in PSM-XL, peak pressure sensitivities were dominated by alignment angle ($S = 0.44$) and collateral stiffness ($S = 0.21$), with friction parameters contributing $S \leq 0.08$ individually. In deep flexion, the posterior translation magnitude had $S = 0.31$ for wear in both models.

We stress-tested robustness by applying simultaneous perturbations at the 95th percentile of distributions. GEX-WEAR gait wear volume varied by $\pm 12\%$ under joint perturbations, while PSM-XL edge peak pressure varied by $\pm 15\%$. For backside micromotion, joint perturbations moved predictions by ± 25 – 30% , underscoring the fragility of inference when penalty stiffness and friction both vary within plausible ranges.

Sensitivity coverage — gradation.

- a. Single-factor local sensitivity checks only
- b. Local sensitivities for multiple inputs without variance apportionment
- c. Local plus Sobol first-order indices for key outputs
- d. Local plus Sobol first-order and total-effect indices, and stress tests under joint perturbations

16. Use History and Use Error Considerations

Both explicit models have been applied in internal design loops and academic studies focused on tibiofemoral wear and malalignment. In prior projects, they supported screening of insert geometries and assessment of laxity effects on contact.

Operator-related pitfalls—such as misassignment of friction coefficients, omission of ligament preloads, or overuse of mass scaling—have been observed occasionally in training contexts and mitigated with checklists and templated input decks. Broader deployment across teams has led to variations in parameter defaults; harmonization efforts are ongoing. These background notes are informative for planning but are not part of the formal grading in this paper.

17. Results: Credibility Scores by Mechanism and Model

We summarize per-factor, per-model, per-mechanism scores in two tables below, one per model. The basis statements in the tables are supported by detailed numbers and comparisons in the Results sections that follow for each mechanism. Composite scores (geometric means) are reported in the narrative for ease of interpretation but are not used to mask factor-level weaknesses.

18. Credibility Table: GEX-WEAR

The following entries enumerate the judgment for each of the eight factors across the four mechanisms for GEX-WEAR. Each basis line is anchored to quantitative evidence described in the Results sections for the corresponding mechanism.

19. Credibility Table: PSM-XL

The following entries enumerate the judgment for each of the eight factors across the four mechanisms for PSM-XL. Each basis line is anchored to quantitative evidence described in the Results sections for the corresponding mechanism.

20. Results: Mechanism 1—Steady-cycle tibiofemoral wear (GEX-WEAR and PSM-XL per-factor scores, 0–12)

GEX-WEAR for steady gait wear received 10/12 for input provenance because 96% of key items (insert lots, serum batches, calibration files, and boundary condition transformations) were linked with identifiers and checksums. The constitutive and wear representation earned 9/12 given the inclusion of viscoelasticity, pressure-dependent plasticity, and a cross-shear wear law calibrated to body-temperature datasets. Defect screening scored 9/12 on the basis of Hertz contact and MMS checks within 2% and wear subroutines passing orthogonal-sliding and restart tests. Discretization and time-integration effects were contained to 1.6% for wear volume and 0.5% for energy drift, hence 9/12. Agreement with experiments was strong: 8% RMS difference in wear volume over 5 Mc across three stations ($N = 6$) and 0.82 ± 0.04 Dice overlap in wear scar maps, scored 9/12. Sensitivity behavior was well characterized—Sobol $S(k_0) = 0.61$ dominated and joint perturbations produced $\pm 12\%$ variation—resulting in 9/12. Quantified uncertainty received 8/12 after 500 LHS samples yielded a 95% predictive interval of [68, 92] mm³ with coverage of 5 of 6 station means. Congruence of boundary conditions with ISO was precise—temperatures

held at 36.5–37.0 °C and AP/IE displacements matching encoder logs—scoring 10/12.

PSM-XL for gait wear earned 9/12 for input provenance, supported by CT-derived alignment, ligament parameters, and instrumented load files referenced with DOIs. The representation scored 8/12 since the ligament springs, while adequate for malalignment studies, introduce approximation for strictly ISO-like gait where the simulator enforces kinematics. Defect screening replicated the benchmark and unit tests to 9/12. Discretization/time-integration effects observed a 2.4% discretization error and 0.6% energy drift at a 0.6 μ s step (8/12). Empirical alignment scored 7/12, reflecting a 14% RMS wear volume error and 0.76 ± 0.06 Dice for scar overlap. Sensitivity behavior ($S(k_0) = 0.57$) and joint perturbation effects of $\pm 14\%$ led to 7/12. Quantified uncertainty, with a [64, 98] mm³ interval capturing 5 of 6 stations, earned 7/12. Boundary mapping from patient telemetry to ISO-like profiles was good, if not exact, scoring 8/12.

Composite scores (geometric mean across the eight factors) were 9.1/12 for GEX-WEAR and 7.8/12 for PSM-XL in this mechanism.

21. Results: Mechanism 2—Edge loading from malalignment/laxity (GEX-WEAR and PSM-XL per-factor scores, 0–12)

For GEX-WEAR under edge loading, input tracking scored 9/12 because malalignment angles, collateral preloads, and sensor calibrations were rigorously documented. The representation obtained 8/12 due to adequate capture of rim contact and plasticity under Stribeck friction. Implementation checks were strong at 9/12, showing stable contact work through sliding reversals. Solver resolution effects warranted 7/12 with peak pressure shifting by 4.2% and 1.4% under successive refinements and reduced kinetic/internal ratios under smaller time steps. Experimental agreement scored 7/12, with peak pressure within 16% and lift-off within 0.10 mm of measurements at 5° valgus. Sensitivity analysis scored 7/12 as alignment angle dominated (Sobol $S(0.44)$) while friction had $S \leq 0.08$. Quantified uncertainty scored 7/12, with 95% intervals covering 9/12 conditions. Reproduction of test boundaries, including synchronized 100 Hz sensing, landed at 8/12.

PSM-XL was particularly strong for edge loading: 9/12 for input tracking, 9/12 for the representation due to its ligament field and explicit management of laxity, and 9/12 for implementation checks. Solver resolution was scored 7/12 based on small but non-zero changes in peak pressure and reduced kinetic/internal energy with smaller steps. Agreement with experiments was 8/12—peak pressure within 12% and lift-off 0.80 mm versus 0.70 ± 0.05 mm. Sensitivity coverage earned 8/12 with alignment angle $S = 0.44$ and ligament stiffness $S = 0.21$. Quantified uncertainty was 7/12: the 95% interval [24, 31] MPa covered 10/12 conditions. Boundary congruence scored 9/12, given the tight control of alignment/laxity combinations and synchronized sensing.

Composite scores were 7.7/12 for GEX-WEAR and 8.2/12 for PSM-XL in edge loading.

22. Results: Mechanism 3—Cross-shear–accelerated wear in deep flexion (GEX-WEAR and PSM-XL per-factor scores, 0–12)

GEX-WEAR received 9/12 for input tracking of torque profiles and rollback targets, 8/12 for the governing representation given the cross-shear law and posterior shift, and 9/12 for implementation tests that specifically exercised cross-shear accumulation. Solver discretization effects scored 7/12 after posterior pressure changed by about 3% with mesh refinement and energy drift remained $\leq 0.6\%$ at 0.5 μs . Agreement with experiments scored 7/12: posterior pressure within 15% and spatial correlation 0.74 for wear maps. Sensitivity behavior received 7/12 with posterior translation magnitude contributing S0.31. Quantified uncertainty scored 7/12 with a 95% wear interval of [15, 29] mm³ at 0.5 Mc and coverage of 4/6 conditions. Boundary mapping to rig logs and flexion excursions earned 8/12.

PSM-XL scored 9/12 for input tracking, 9/12 for the representation due to ligament field effects on posterior shift and sliding, and 9/12 for implementation tests. Solver effects landed at 7/12 with small residual changes in pressure at finer meshes. Agreement with experiments scored 8/12 (posterior pressure within 15% and spatial correlation 0.78). Sensitivity behavior was 7/12, as in GEX-WEAR, and quantified uncertainty was 7/12 with the interval [16, 31] mm³. Congruence with boundary conditions, including mapping to telemetry loads for combined loading in deep flexion, earned 9/12.

Composite scores were 7.7/12 (GEX-WEAR) and 8.0/12 (PSM-XL) for deep flexion pivot wear.

23. Results: Mechanism 4—Backside fretting at insert–baseplate interface (GEX-WEAR and PSM-XL per-factor scores, 0–12)

GEX-WEAR backside fretting scored 8/12 for input tracking with optical sensor calibrations and versioned load sets from two insert lots. The representation earned 7/12 as it captured Coulomb friction and conforming contact but not micro-topography or third-body effects. Implementation checks were strong (9/12), including analog transition thresholds within 5%. Solver effects scored 6/12 due to sensitivity of micromotion to penalty tuning and a numerical jitter contribution of roughly 10–15% of variance. Agreement with experiments scored 6/12: micromotion within a factor of 1.2–1.8 and fretting power within 30% at midrange but deviating up to 45% at low preload. Sensitivity behavior scored 6/12—joint perturbations shifted micromotion by ± 25 –30%. Quantified uncertainty scored 5/12 with broad 95% intervals [22, 58] μm and coverage in 6/8 conditions. Boundary congruence scored 6/12; while axial and shear loads matched, the rig could not replicate complex in vivo multiaxial coupling.

PSM-XL backside fretting scored 8/12 for input tracking. The representation was less specialized (6/12), with no explicit micro-topography and without tuning for fretting. Implementation checks scored 9/12. Solver effects scored 5/12 due to 15–25% micromotion variation across penalty and time-step

choices. Empirical agreement scored 5/12 (micromotion within a factor of 1.5–2.1 and fretting power differences up to 45% at low preload). Sensitivity behavior scored 5/12 with strong friction–penalty interaction. Quantified uncertainty scored 5/12 with a [26, 64] μm interval and 5/8 coverage. Boundary congruence was 5/12 owing to limited multi-axial coupling in the rig relative to plausibly coupled in vivo conditions.

Composite scores were 6.5/12 (GEX-WEAR) and 5.9/12 (PSM-XL) for backside fretting.

24. Synthesis of Findings Across Models and Mechanisms

A single summary statement per credibility dimension helps interpret cross-cutting lessons while the detailed rows in the tables preserve mechanism-specific judgments. Regarding the chain-of-custody for inputs and calibrations, we observed near-complete coverage across models and tests, with 96% of items linked to traceable records; this depth of documentation reduces ambiguity when reconciling minor discrepancies across laboratories and rigs. For the governing representation, the explicit solvers, nonlinear UHMWPE with pressure-sensitive plasticity, and cross-shear wear formulation jointly captured the salient physics across mechanisms; residual limitations stem from approximated lubrication and lack of explicit micro-topography, particularly for backside fretting. Implementation checks were extensive and passed with tight tolerances, with MMS, Hertz contact, cross-shear unit tests, and regression suites curtailing the risk of sign or logic errors. Discretization and step-size effects were small for gait and moderate for edge and pivot conditions; backside micromotion was more sensitive to penalty and time step choices, which complicates inference in regimes of very small slips. Agreement with experiments was strong where ISO-like and well-instrumented tests constrained loads and kinematics; it weakened where test conditions were sparse or less physiological, as in backside fretting. Sensitivity behavior was well-mapped for gait and edge loading and illuminated which parameters are decision-critical (e.g., k_0 and alignment angles). Uncertainty quantification captured most observed results within 95% predictive bands, with broader intervals in fretting that mirrored test–model variability. Finally, the match between simulated boundary conditions and those in the tests was excellent for ISO gait and tightly controlled for malalignment/laxity and deep flexion but inevitably less faithful for backside fretting, where in vivo-like multi-axial coupling is challenging to reproduce in a bench fixture.

Cross-cutting synthesis quality — gradation.

- a. Lists of results without integration
- b. Qualitative integration without mechanism or factor distinctions
- c. Mechanism-specific synthesis by factor with explicit trade-offs
- d. Mechanism- and factor-specific synthesis with articulated trade-offs and implications for decision-making

25. Planned Elevation Strategies by Factor

To raise confidence for backside fretting, we will expand bench measurements to include multi-axial coupling with superposed mediolateral moments and IE torques derived from telemetry, and we will incorporate micro-topography via an asperity-informed friction model or an effective roughness parameter calibrated to new tests. For deep flexion pivot, we aim to instrument posterior contact regions with thin, high-durability sensors capable of higher flexion without wrinkling to cross-validate pressure films.

For solver resolution effects, we will explore adaptive contact refinement and variable mass scaling keyed to contact stiffness to reduce numerical jitter without inflating run times. We will adopt contact stabilization schemes that suppress high-frequency chatter without damping out physical slip. On the uncertainty side, we will add hierarchical models for k_0 and β to incorporate lot-to-lot variation directly and to encode prior knowledge from pin-on-plate studies.

To strengthen empirical alignment, particularly for edge and pivot mechanisms, we plan joint inter-laboratory studies that replicate our malalignment and deep flexion rigs with independent datasets and blinded parameter settings. We will publish input decks and parameter priors with persistent identifiers to enable external reproduction and challenge studies.

In PSM-XL, ligament representation will be refined by integrating subject-specific stiffness profiles when preoperative MRI delineates attachments and by constraining spring curves from cadaver tests spanning a wider age and tissue quality range. For both models, we will document and automate more of the input assembly process to further reduce operator variability.

26. Limitations

This assessment is limited by the available datasets. ISO-level gait is richly characterized and affords precise comparisons; deep flexion pivot and backside fretting are less standardized, leading to broader predictive intervals and more conservative scores. We did not resolve mixed lubrication or cavitation phenomena that may influence friction at very low sliding speeds in deep flexion. The backside interface simulations used nominally conforming surfaces and Coulomb friction without explicit debris, which constrains interpretation under third-body conditions. Ligament representation through springs reproduces gross laxity but not the detailed attachment geometry or anisotropy of ligament tissue; nonetheless, it captures the key load–deflection features relevant to edge loading and pivoting maneuvers.

While we implemented extensive numerical checks, the explicit solvers necessarily employ mass scaling and penalty contact; although tuned conservatively, these choices can still interact with parameter uncertainty under small-slip fretting conditions. Retrieval comparisons provided qualitative pattern checks but did not furnish individualized load histories, limiting the ability to make subject-specific claims. Finally, we did not assign formal grades for operator misuse or historical

deployment patterns, though both inform practical application risks.

27. Conclusions

The two explicit FEA models evaluated here—GEX-WEAR and PSM-XL—demonstrate credible performance for tibiofemoral wear under gait and for edge loading and deep flexion pivot phenomena in a posterior-stabilized TKR. GEX-WEAR achieved a composite of 9.1/12 for steady-cycle gait wear with close agreement to ISO simulator tests, and PSM-XL excelled in edge loading with a composite of 8.2/12 and in deep flexion pivot with 8.0/12. Both models are informative but not definitive for backside fretting given sensitivity to penalty settings and limited bench data; composites were 6.5/12 (GEX-WEAR) and 5.9/12 (PSM-XL). The explicit solvers, UHMWPE constitutive choices, and cross-shear wear law form a sound basis for preclinical evaluation when paired with rigorous numerical discipline and carefully matched test conditions.

The mechanism-specific scoring clarifies where each model is strongest and where further evidence is needed. For regulatory or design decisions centered on steady-state wear or edge-induced damage, the presented evidence supports use of these models as part of a convergent evidence package. For backside fretting, additional targeted experiments and solver refinements are necessary to elevate confidence. The per-factor tables, together with the procedures and numbers presented, are intended to facilitate transparent review and reproducible improvements under 7009A.

28. Supplemental Material (input datasets, element quality checks, scoring tables)

Supplemental files include: (1) CSVs of input parameter priors and distributions for uncertainty analyses; (2) element quality histograms (skewness, aspect ratio) for tibiofemoral and backside contact regions for both models; (3) MMS and Hertz benchmark scripts and outputs; (4) detailed scoring spreadsheets that mirror the tables printed in this paper with additional reviewer notes; and (5) processing notebooks for converting instrumented load datasets to simulation boundary conditions.

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Table 1: Per-factor credibility scores for GEX-WEAR across four mechanisms (0–12 scale).

Credibility factor	Level	Basis
Data pedigree - GEX-WEAR - Steady-cycle tibiofemoral surface wear under gait	10/12	96% of inputs linked to lot numbers and calibration files; ISO deck and serum logs versioned with checksums
Model form - GEX-WEAR - Steady-cycle tibiofemoral surface wear under gait	9/12	Nonlinear UHMWPE with viscoelasticity and pressure-dependent plasticity plus cross-shear wear; lubrication via Stribeck
Numerical code verification - GEX-WEAR - Steady-cycle tibiofemoral surface wear under gait	9/12	Hertz and MMS checks within 1.5–2%; wear subroutines passed cross-shear and restart integrity tests
Numerical solver error - GEX-WEAR - Steady-cycle tibiofemoral surface wear under gait	9/12	Mesh/time-step studies yielded $\leq 1.6\%$ discretization error and $\leq 0.5\%$ energy drift over 10,000 cycles
Output comparison - GEX-WEAR - Steady-cycle tibiofemoral surface wear under gait	9/12	Wear volume within 8% RMS at 5 Mc (N = 6); scar Dice overlap 0.82 ± 0.04
Results robustness - GEX-WEAR - Steady-cycle tibiofemoral surface wear under gait	9/12	Sobol S(k0)=0.61; joint perturbations varied wear by $\pm 12\%$; mesh changes altered wear by 1.1%
Results uncertainty - GEX-WEAR - Steady-cycle tibiofemoral surface wear under gait	8/12	95% predictive interval [68, 92] mm ³ captured 5/6 station means; inputs sampled N=500 LHS
Test conditions - GEX-WEAR - Steady-cycle tibiofemoral surface wear under gait	10/12	ISO 14243-3 profiles reproduced; temperature 36.5–37.0 °C; AP/IE displacements matched encoder logs
Data pedigree - GEX-WEAR - Edge loading from varus–valgus malalignment and laxity	9/12	Malalignment angles and spring preloads tracked; Tekscan calibrations recorded with timestamps
Model form - GEX-WEAR - Edge loading from varus–valgus malalignment and laxity	8/12	Penalty contact with Stribeck friction and UHMWPE plasticity captured rim contact and shear
Numerical code verification - GEX-WEAR - Edge loading from varus–valgus malalignment and laxity	9/12	Contact benchmarks and sign tests showed stable, non-negative contact work under edge transitions
Numerical solver error - GEX-WEAR - Edge loading from varus–valgus malalignment and	7/12	Refinement (0.60.40.3 mm) shifted peak pressure by 4.2% and 1.4%; time-step cuts lowered kinetic ratios

Table 2: Per-factor credibility scores for PSM-XL across four mechanisms (0–12 scale).

Credibility factor	Level	Basis
Data pedigree - PSM-XL - Steady-cycle tibiofemoral surface wear under gait	9/12	CT-derived alignment, ligament parameters, and instrumented loads linked to DOIs; wear simulator lots tracked
Model form - PSM-XL - Steady-cycle tibiofemoral surface wear under gait	8/12	UHMWPE nonlinearity and cross-shear wear included; ligament springs approximated for gait loads
Numerical code verification - PSM-XL - Steady-cycle tibiofemoral surface wear under gait	9/12	MMS, Hertz, and wear unit tests matched benchmarks within 2%; nightly regression checks
Numerical solver error - PSM-XL - Steady-cycle tibiofemoral surface wear under gait	8/12	Discretization error 2.4% at finest practical mesh; energy drift 0.6% at 0.6 μ s
Output comparison - PSM-XL - Steady-cycle tibiofemoral surface wear under gait	7/12	Wear volume within 14% RMS; scar Dice 0.76 ± 0.06
Results robustness - PSM-XL - Steady-cycle tibiofemoral surface wear under gait	7/12	Sobol S(k0)=0.57; joint perturbations varied wear by $\pm 14\%$
Results uncertainty - PSM-XL - Steady-cycle tibiofemoral surface wear under gait	7/12	95% predictive interval [64, 98] mm ³ ; 5/6 station means inside interval
Test conditions - PSM-XL - Steady-cycle tibiofemoral surface wear under gait	8/12	ISO-like kinematics mapped from patient loads; temperature and serum controlled
Data pedigree - PSM-XL - Edge loading from varus–valgus malalignment and laxity	9/12	Alignment and laxity tuning files versioned; Tekscan and LVDT calibrations tied to trials
Model form - PSM-XL - Edge loading from varus–valgus malalignment and laxity	9/12	Explicit penalty contact with ligament spring field reproduced laxity-affected edge behavior
Numerical code verification - PSM-XL - Edge loading from varus–valgus malalignment and laxity	9/12	Contact benchmarks and edge-transition tests passed; contact work positive definite
Numerical solver error - PSM-XL - Edge loading from varus–valgus malalignment and laxity	7/12	Refinement to 0.3–0.4 mm changed peak pressure by 1.6%; time-step cuts reduced kinetic/internal to 0.03
Output comparison - PSM-XL - Edge	8/12	Peak pressure within 12%; lift-off 0.80 mm vs 0.70 $\pm$